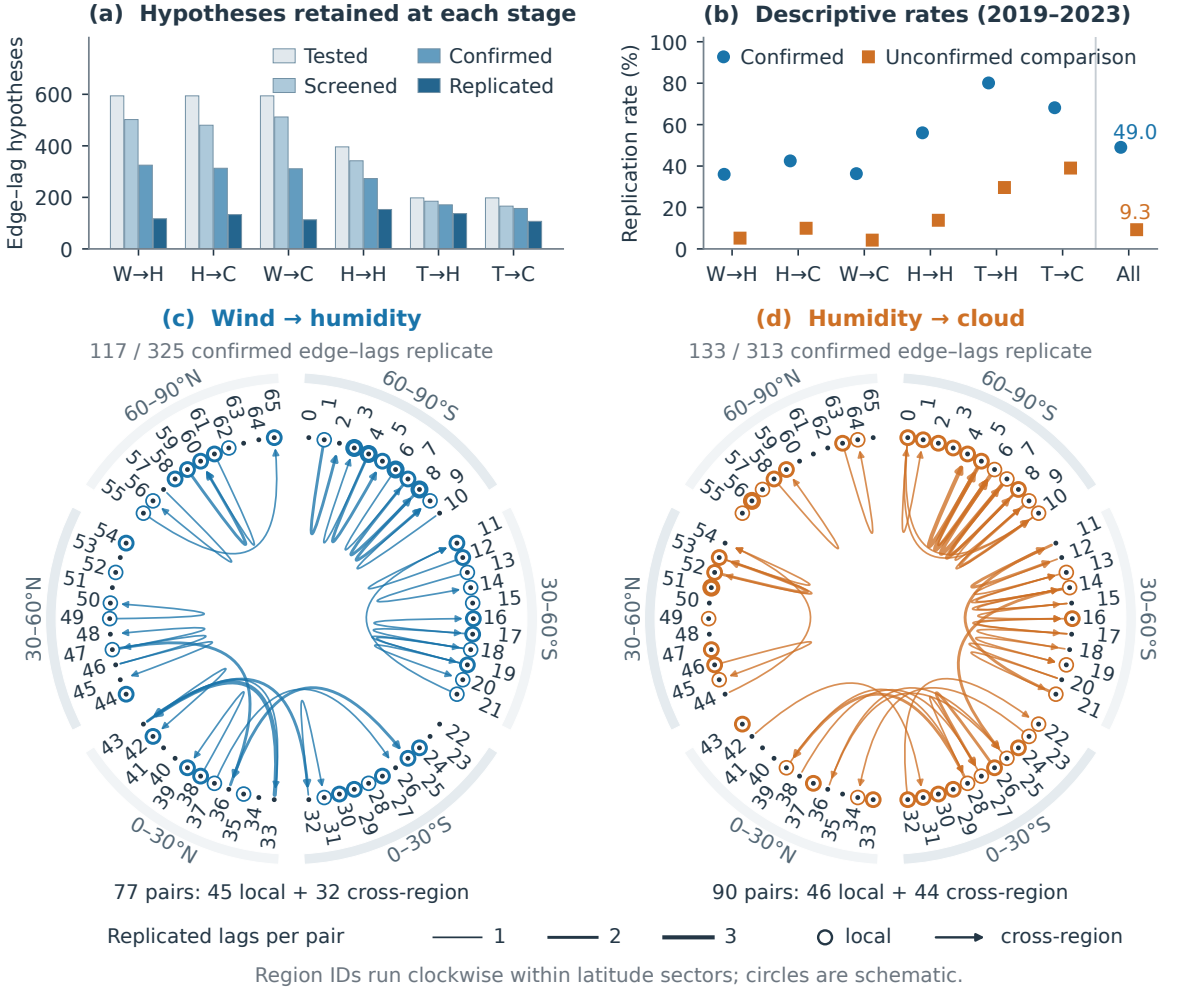

(a) Hypotheses retained at each stage
Edge–lag hypotheses
Tested
Screened
Confirmed
Replicated
0
200
400
600
W→H
H→C
W→C
H→H
T→H
T→C
(b) Descriptive rates (2019–2023)
Replication rate (%)
Confirmed
Unconfirmed comparison
0
20
40
60
80
100
49.0
9.3
W→H
H→C
W→C
H→H
T→H
T→C
All
(c) Wind → humidity
117 / 325 confirmed edge–lags replicate
60–90°N
60–90°S
30–60°N
30–60°S
0–30°N
0–30°S
77 pairs: 45 local + 32 cross-region
(d) Humidity → cloud
133 / 313 confirmed edge–lags replicate
60–90°N
60–90°S
30–60°N
30–60°S
0–30°N
0–30°S
90 pairs: 46 local + 44 cross-region
Replicated lags per pair
1
2
3
local
cross-region
Region IDs run clockwise within latitude sectors; circles are schematic.